STATS 280 - Midterm 2

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Q.1 (30)

Q.1 a

(6 points) There are 18 participants and 3 traitors, so that the chance is $3/18 = 1/6$.

Q.1 b

(6 points) The chance that the first person banished is a faithful is 5/6 from part a., the chance that the second person banished is a faithful is 13/16 since there are 3 traitors and 16 participants at that point, and the chance that the third person banished is a faithful is 11/14. Hence the chance that the first 6 participants removed from the game are all faithfuls is $\frac{5}{6}\frac{13}{16}\frac{11}{14} = \frac{715}{1{,}344} = 0.53$.

Q.1 c

(6 points) This is the opposite of the event in b., and so the chance is $1 - 0.53 = 0.47$

Q.1 d

(12 points) The first possibility is that the first two banished participants are traitors and the third is a faithful. This chance is $\frac{1}{6}\frac{2}{16}\frac{12}{14} = 0.018$ since one faithful would become a traitor after the second banishment. The second possibility is that the first banished participant is a faithful and the next two are traitors. This has chance $\frac{5}{6}\frac{3}{16}\frac{2}{14} = 0.022$. The third possibility is that the first and third participants banished are traitors and the second participant banished a faithful. This has chance $\frac{1}{6}\frac{14}{16}\frac{2}{14} = 0.021$. Hence the total chance is $0.018 + 0.022 + 0.021 = 0.061$.

Q.2 (30)

Q.2 a

(8 points) The maximum possible overlap will be the size of the smallest subset, which is 20 for the number of employees of Asian descent. This overlap would occur when every employee of Asian descent is under 30 years of age, every employee under 30 years of age is a woman, 20 employees under 30 years of age are not of Asian descent, 10 employees are women but not of Asian descent and not under 30 years of age, and 50 employees are neither women nor under 30 years of age nor of Asian descent.

Q.2 b

(8 points) Yes, this is possible. One possibility is where all 40 employees who are under 30 years of age are women, 10 women are not under 30 years of age, none of the 20 employees of Asian descent are women, and 30 employees are not women, not under 30 years of age, and not of Asian descent.

Q.2 c

(7 points) We know that $P(woman\ or\ under\ 30) = P(woman) + P(under\ 30) - P(woman\ and\ under\ 30) = 0.5 + 0.4 - 0.27 = 0.63$. This means that there are 63 employees who are women or under 30 years of age.

Q.2 d

(7 points) If being a woman and being under 30 years of age are independent, then $P(woman\ and\ under\ 30) = P(woman)P(under\ 30) = 0.5 * 0.4 = 0.2$. This would imply that there are 20 employees that are women under 30 years of age. The number in part c) is greater than what we would expect if being a woman and being under 30 years of age were independent.

Q.3 (40)

A tree diagram for this problem will have the year at the first level, with weights $0.40, 0.35\ and\ 0.25$ for first year, second year, and third year students, respectively. The second level of the tree will have study times, with fewer than 10 hours, at least 10 hours and less than 20 hours, and at least 20 hours weights of $0.50, 0.35\ and\ 0.15$ for first year students, $0.30, 0.45\ and\ 0.25$ for second year students and $0.20, 0.40\ and\ 0.40$ for third year students, respectively.

Q.3a

(10 points) This probability is $0.40 * 0.15 + 0.35 * 0.25 + 0.25 * 0.4 = 0.25$ by the law of total probability.

Q.3b

(10 points) This probability will be

$$P(1st\ year|at\ least\ 20) = P(1st\ year\ and\ at\ least\ 20)/P(at\ least\ 20) \tag{1}$$
$$= \frac{0.40 * 0.15}{0.25} \tag{2}$$
$$= 0.24 \tag{3}$$

Q.3c

(10 points) This probability will be

$$P(2nd\ year|at\ least\ 20) = P(2nd\ year\ and\ at\ least\ 20)/P(at\ least\ 20) \tag{4}$$
$$= \frac{0.35 * 0.25}{0.25} \tag{5}$$
$$= 0.35 \tag{6}$$

Q.3d

(10 points) This probability will be

$$P(3rd\ year|at\ least\ 20) = P(3rd\ year\ and\ at\ least\ 20)/P(at\ least\ 20) \tag{7}$$
$$= \frac{0.25 * 0.4}{0.25} \tag{8}$$
$$= 0.4 \tag{9}$$

Q.4 (30)

Q.4a

(8 points) This is the hypergeometric probability $\binom{12}{2}\binom{8}{3}/\binom{20}{5}$. Simplification gives $\binom{12}{2} = 12 * 11/2 = 66$, $\binom{8}{3} = 8 * 7 * 6/6 = 56$, and $\binom{20}{5} = 20 * 19 * 18 * 17 * 16/(5 * 4 * 3 * 2 * 1) = 15,504$. Hence the chance is $66 * 56/15,504 = 0.24$.

Q.4b

(5 points) The command will be dhyper(2,12,8,5).

Q.4c

(12 points) This is $1 - P(2\ or\ fewer\ new)$. We can get the chances from the binomial distribution as $P(0\ new) = (8/20)^5 = 0.01$, $P(1\ new) = 5 * (12/20) * (8/20)^4 = 0.077$, $P(2\ new) = \binom{5}{2} * (12/20)^2 * (8/20)^3 = 0.23$. Hence the chance is $1 - (0.01 + 0.077 + 0.23) = 0.68$.

Q.4d

(5 points) The command will be 1-pbinom(2,5,0.6).

Q.5 (40)

Q5.a

(20 points) The table below summarizes the possibilities for the die roll N, tickets T, and maximum M

n	P(N=n)	n+1	T	M
0	1/8	1	$\{1\},\{1\},\{2\},\{4\}$	$1,1,2,4$
1	3/8	2	$\{1,1\},\{1,2\},\{1,4\},\{1,2\},\{1,4\},\{2,4\}$	$1,2,4,2,4,4$
2	3/8	3	$\{1,1,2\},\{1,1,4\},\{1,2,4\},\{1,2,4\}$	$2,4,4,4$
3	1/8	4	$\{1,2,3,4\}$	4

Compiling the possibilities for M from the table above gives the table below.

m	P(M=m)
1	$1/8 * 1/2 + 3/8 * 1/6 = 1/8$
2	$1/8 * 1/4 + 3/8 * 2/6 + 3/8 * 1/4 = 1/4$
4	$1/8 * 1/4 + 3/8 * 1/2 + 3/8 * 3/4 + 1/8 = 5/8$

Q.5 b

(7 points) The expected value will be $E(M) = 1 * 1/8 + 2 * 1/4 + 4 * 5/8 = 25/8$.

Q.5 c

(8 points) Tossing the coin is like drawing with replacement from the box $\{0,1\}$, and the number of heads is the sum. The following code gives one way to simulate the random experiment above

```
simM=function() {
    numDraws = sum(sample(c(0,1),3,replace=TRUE))+1
    tickets = sample(c(1,1,2,4),numDraws,replace=FALSE)
    return(max(tickets))
    }
```

Q.5 d

(5 points) This code is estimating the standard error of M.

Q.6 (30)

Q.6 a

(5 points) The R command would be pbinom(100,150,0.5)-pbinom(69,150,0.5)

Q.6 b

(15 points) The continuity correction gives 69.5 to 100.5 balls. The expected number of reds is $150 * 1/2 = 75$ and the standard error is $\sqrt{0.5 * 0.5 * 150} = 6.12$. Converting to standard units gives $z_1 = (69.5 - 75)/6.12 = -0.90$ and $z_2 = (100.5 - 75)/6.12 = 4.17$. The table in the back of the book gives roughly $0.632/2 + 0.5 = 0.82$.

Q.6 c

(5 points) The standard units are 4.17 from part c. and so the chance is less than $(1 - 0.999967)/2 = 0.000017$ from the table in the back of the book.

Q.6 d

(5 points) Picking 100 or more red balls is very unlikely if my friend is telling the truth. I do not believe that my friend is telling the truth.